RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Third Year – Semester II Examination – March-May 2025
MAT 3222 – NUMERICAL ANALYSIS II

Time: Two (02) hours

Answer **ALL** (04) questions.
Use of a calculator is permitted.
Perform all calculations with accuracy up to four decimal places.

1. a) Define partial pivoting and explain its necessity in Gaussian elimination. **(10 marks)**

   b) Solve the following system of linear equations using Gaussian elimination method:

$$\begin{cases} 0.02x_1 + 1.0x_2 + 0.5x_3 = 2.1 \\ 0.5x_1 + 0.2x_2 + 0.01x_3 = 1.8 \\ 0.1x_1 - 0.1x_2 + 0.3x_3 = 0.7. \end{cases}$$

   (40 marks)

   c) State the Existence and Uniqueness Theorem for the Fixed-Point Method.

   (05 marks)

   d) Consider the nonlinear equation: $x^3 + x - 1 = 0$. Let consider the fixed-point iteration function: $g(x) = x - \frac{x^3+x-1}{3x^2+1}$. Assume that the root lies in the interval [0,1] and take the initial approximation as $x_0 = 0.5$.

      i. Check whether the function $g(x)$ satisfies the convergence condition for fixed-point iteration in the interval [0,1]. **(10 marks)**

      ii. If the method is convergent, apply the fixed-point iteration method to obtain the root correct to the tolerance 10^{-3}. **(35 marks)**

2. a) Compare Newton's method and Broyden's method for solving nonlinear equations, highlighting their main differences. **(20 marks)**

   b) Consider the nonlinear system:

$$\begin{cases} f_1(x_1, x_2, x_3) = {x_1}^2 + {x_2}^2 - 4 = 0 \\ f_2(x_1, x_2, x_3) = x_1x_2 + x_1 - 1 = 0 \\ f_3(x_1, x_2, x_3) = {x_2}^2 + {x_3}^2 - 5 = 0, \end{cases}$$

Cont.

with initial guess: $x^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

i. Use Newton's method to compute the first approximation $x^{(1)}$. **(30 marks)**

ii. Using the result from Part i, apply Broyden's method to compute the second approximation $x^{(2)}$. **(50 marks)**

3. a) i. State the Fourth-Order Runge–Kutta Method. **(05 marks)**

ii. Consider the following initial value problem:

$$\frac{dy(t)}{dt} = y(t) - t^2 + 1; \quad t \in [0,1]$$

with initial condition $y(0) = 0.75$.

Using the Fourth-Order Runge–Kutta Method with step size $h = 0.25$, approximate $y(0.50)$. **(35 marks)**

b) Consider the third-order Ordinary Differential Equation:

$$y'''(t) = t^2 + y''(t) - y'(t) + y(t)$$

with the initial conditions;

$$y(0) = 1, y'(0) = 2, y''(0) = 0 \text{ and } t \in [0,1].$$

The exact solution of the differential equation:

$$y(t) = 1.5e^t - 0.5cost + 2.5sint - t^2 - 2t.$$

i. Reformulate the given higher-order differential equation as a system of first-order equations in vector (matrix) form. Using the third-order Taylor series method based on higher-order derivatives, compute an approximation to $y(0.3)$ for $h = 0.15$. **(50 marks)**

ii. It is found that $y(0.3) \approx 1.5960$ for a step size of $h = 0.10$.

Using the exact solution, compare this result with the result from Part i, and comment on how the choice of step size h influences the accuracy of the Taylor series method. **(10 marks)**

4. Let $x_i = a + ih\,;\ i = 0,1,\ldots,N+1,$

where $h = \frac{(b-a)}{N}$.

Consider the second–order Boundary Value Problem:

$$y'' = p(x)y' + q(x)y + r(x); \quad a \le x \le b,$$

with boundary conditions:

$$y(a) = \alpha, y(b) = \beta.$$

a) Using the Linear Shooting Method, show that the solution of the above boundary value problem can be written in the form:

$$y(x) = y_1(x) + Cy_2(x),$$

where $y_1(x)$ and $y_2(x)$ are solutions of appropriate initial value problems and C is a constant to be determined from the boundary conditions. **(30 marks)**

b) Using the Linear Finite Difference Method, derive a system of linear equations used to approximate the solution of the boundary value problem. **(40 marks)**

c) Use the Linear Finite Difference Method with step size, $h = \frac{1}{4}$ to approximate the solution of the boundary value problem:

$$y''(x) - 2y'(x) - 3y(x) = x^2; \qquad 0 \leq x \leq 1$$

subject to,

$$y(0) = 1,\ y(1) = 2.$$

(30 marks)

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